

11__While__Loop

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```
[2]: # To perform a task multiple times for a given set of inputs or entries then we
      ↳ will go for loops (While loop, For loop)
      # In While loop, the body will be continuously executed until the condition is
      ↳ true
      # Iteration over each input/entry should be stopped at one point either by
      ↳ updating the iterator in the loop or condition failure at one point
```

```
counter = 1
while counter < 10:
    counter += 1
    print("The Number updated to ",counter)
```

```
The Number updated to 2
The Number updated to 3
The Number updated to 4
The Number updated to 5
The Number updated to 6
The Number updated to 7
The Number updated to 8
The Number updated to 9
The Number updated to 10
```

```
[4]: # Nested while loop example
teacher_cnt = 1
principal_cnt = 1

while teacher_cnt <= 3:
    print("Teacher's Turn ",teacher_cnt)
    while principal_cnt <= 3:
        print("--> Principal's Turn ",principal_cnt)
        principal_cnt += 1
    teacher_cnt += 1
```

```
Teacher's Turn 1
--> Principal's Turn 1
--> Principal's Turn 2
--> Principal's Turn 3
Teacher's Turn 2
Teacher's Turn 3
```

```
[7]: num_list = [2,3,5,7,10]
sum_sqaure_nums= 0
index=0

while index < len(num_list):
    sum_sqaure_nums += num_list[index]**2
    index += 1

print("Sum of squares of all given numbers is ", sum_sqaure_nums)
```

Sum of squares of all given numbers is 187

```
[8]: # While loop Else

teacher_cnt = 1
principal_cnt = 1

while teacher_cnt <= 3:
    print("Teacher's Turn ",teacher_cnt)
    while principal_cnt <= 3:
        print("--> Principal's Turn ",principal_cnt)
        principal_cnt += 1
    else:
        print("--> Principal's Turn already completed")
    teacher_cnt += 1
```

```
Teacher's Turn 1
--> Principal's Turn 1
--> Principal's Turn 2
--> Principal's Turn 3
--> Principal's Turn already completed
Teacher's Turn 2
--> Principal's Turn already completed
Teacher's Turn 3
--> Principal's Turn already completed
```